



Intro to Computer Networks

Brief Overview

This note covering **Computer Networking** was created from an [133-page uploaded document](#). It walks you through fundamental concepts such as network basics, Internet architecture, key protocols, performance factors, and common networking devices.

Key Points

- Core definitions of networks, hosts, and the Internet.
 - Overview of access and core network technologies (wired, wireless, fiber).
 - Explanation of delay, loss, throughput, and queuing in packet-switched networks.
 - Summary of typical network devices and their roles in a modern LAN/WAN.
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Introduction to Computer Networks

Computer network: *Two or more devices connected together that communicate to share data or resources.*

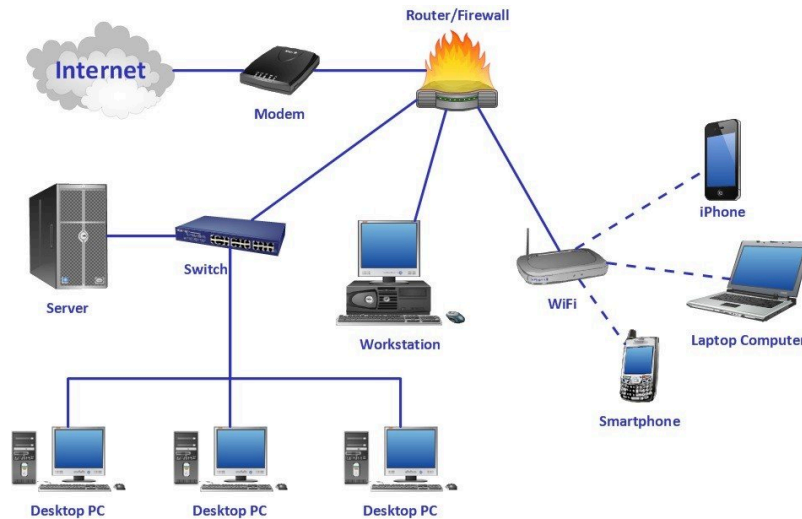
- Networks enable **resource sharing** (files, printers, internet access).
 - Devices may be **end systems** (hosts) or **network infrastructure** (routers, switches).
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What is the Internet?

Internet: *A massive “network of networks” that interconnects billions of computing devices worldwide.*

- **End systems** include PCs, servers, smartphones, TVs, appliances, wearables, cars, etc.
- **Network components:** packet switches (routers, switches), communication links (fiber, copper, radio, satellite).
- **Transmission rate** of a link is its **bandwidth** (bits per second).



The diagram illustrates the typical hierarchy from the Internet down to end devices.

Internet Standards & Protocols

- **RFC** – *Request for Comments*, the formal document series for Internet standards.
- **IETF** – *Internet Engineering Task Force*, the organization that develops RFCs.

Common protocols (non-exhaustive):

Layer (TCP/IP)	Example Protocols
Application	HTTP, FTP, SMTP, DNS
Transport	TCP, UDP
Network	IP
Link	Ethernet, Wi-Fi, 4G/5G

Protocols define **message formats**, **ordering**, and **actions** for sending/receiving data.

☒ Network Edge

Hosts – clients & servers

Host: *A device that runs application programs (e.g., web browser, email client, web server).*

- **Clients** request services; **servers** provide them.
- Servers are often located in data-center **core** networks.

Access Networks & Physical Media

Wired access

- **DSL** – uses existing telephone lines; splitter separates voice and data.
 - Downstream: 24 kHz–1 MHz ($\approx$ 24–52 Mbps)
 - Upstream: 4 kHz–50 kHz ($\approx$ 3.5–16 Mbps) – **asymmetric**.
- **Cable (HFC)** – hybrid fiber-coax; shared medium with CMTS at the headend.
 - Downstream up to 1.2 Gbps, upstream 30–100 Mbps.
- **Ethernet** – common in homes & enterprises; speeds 100 Mbps, 1 Gbps, 10 Gbps.

Wireless access

- **Wi-Fi (802.11b/g/n)** – typical indoor range $\sim$ 100 ft; rates 11, 54, 450 Mbps.
- **Cellular (4G/5G)** – wide-area coverage; tens of Mbps, future 6G.
- **Satellite & microwave** – long-distance links; up to 45 Mbps, higher latency.

Physical media types

Media	Characteristics
Twisted pair (TP)	Copper pairs, UTP/STP; Cat-5 $\rightarrow$ 100 Mbps/1 Gbps, Cat-6 $\rightarrow$ 10 Gbps
Coaxial cable	Two concentric conductors; broadband, 100s of Mbps per channel
Fiber optic	Light pulses in glass; 10-100 Gbps, low error, immune to EM noise
Wireless radio	Free-space propagation; subject to reflection, obstruction, interference



Fiber offers high-speed, low-error transmission.

Example IoT devices (fun inter-connected items)





Smart speakers, toasters, cameras, and wearables illustrate the expanding “Internet of Things.”

Network Core

Packet Switching & Store-and-Forward

- **Transmission delay** for a packet of length L bits on a link of rate R (bps):
Transmission delay = $\frac{L}{R}$ seconds
- **Store-and-forward**: a router must receive the entire packet before forwarding it.
- **One-hop example**: $L = 10, \text{Kbits}$, $R = 100, \text{Mbps}$ $\rightarrow$ delay = 0.1,ms.

Queuing, Loss & Congestion

- When **arrival rate** $> R$, packets queue in the router buffer.
- If the buffer fills, **packet loss** occurs.

Forwarding vs. Routing

- **Forwarding** – local action: move a packet from an input link to the appropriate output link based on its header.
- **Routing** – global action: compute the path (source-destination) for packets; updates routing tables.

Circuit Switching

- Dedicated **circuits** are reserved for the duration of a call.
- **FDM** – each call gets its own frequency band.
- **TDM** – time is divided into slots; each call uses a periodic slot.

Comparison: Packet vs. Circuit Switching

Feature	Packet Switching	Circuit Switching
Connection type	Connectionless	Connection-oriented

Typical traffic	Data (bursty)	Voice (continuous)
Flexibility	High	Inflexible
Ordering	May arrive out-of-order, reassembled at destination	Preserves order
Switching method	Store-and-forward	FDM / TDM
Layer	Network layer (IP)	Physical layer (circuit)
Bandwidth usage	Dynamic (efficient)	Fixed (often under-utilized)
Delay components	Transmission delay + possible queuing	Call-setup delay + constant transmission

Numerical example (circuit-switched)

- File size: 640,000 bits
- Link rate: 1.536 Mbps, 24-slot TDM → per-circuit rate = 64 kbps
- Transmission time: $640,000 / 64,000 = 10, \text{s}$
- Add 0.5 s setup → **10.5 s total**.

Internet Structure – A “Network of Networks”

- **Access ISPs** connect end systems to the Internet (residential, enterprise, university).
- **Transit ISPs** interconnect many access ISPs; **Tier 1 ISPs** have global reach without paying other ISPs.
- **IXPs (Internet Exchange Points)** allow multiple ISPs to peer directly, reducing latency and cost.
- **Content provider networks** (e.g., Google, Microsoft) operate their own backbone to place services close to users.

Typical hierarchy

1. **End systems** → Access ISP
2. **Regional ISP** → Peering at IXPs
3. **Tier 1 ISP** → Global backbone

Performance: Delay, Loss & Throughput

- **Delay** sources: transmission, propagation, processing, queuing.
- **Loss** occurs when queues overflow.
- **Throughput** = effective data rate achieved by a flow; limited by the **bottleneck link** (minimum R along the path).

Understanding these metrics guides the design of **congestion-control** and **reliable transport** protocols.



Packet Delay Sources

Processing delay (d_{proc}): Time a router spends examining a packet's header, checking for bit errors, and determining the outgoing link. Typically **< 1 μ s**.

Queuing delay (d_{queue}): Time a packet waits in the router's buffer before it can be transmitted. Depends on current traffic load; ranges from **μ s to ms**.

Transmission delay (d_{trans}):

$$d_{trans} = \frac{L}{R}$$

where L = packet length (bits) and R = link transmission rate (bps).

Propagation delay (d_{prop}):

$$d_{prop} = \frac{d}{s}$$

where d = physical length of the link and $s \approx 2 \times 10^8$, m/s (speed of signal in the medium).

Overall nodal delay:

$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$



Queuing Delay & Traffic Intensity

Traffic intensity (I): Ratio of the average bit arrival rate to the link's transmission rate.

$$I = \frac{L_a}{R} = \frac{L \cdot a}{R}$$

where a = average packet arrival rate (packets/s) and $L_a = L \cdot a$ (bits/s).

- **If $I > 1$** $\rightarrow$ arrival rate exceeds service rate $\rightarrow$ **average queuing delay $\rightarrow \infty$** (persistent congestion).

- If $I \leq 1 \rightarrow$ system can keep up; queuing delay grows as $I \rightarrow 1$.

Approximate queuing delay for an $M/D/1$ queue (when $I < 1$):

$$d_{queue} \approx \frac{I, L}{R, (1-I)}$$

Design guideline: **Keep** $I \leq 1$ to avoid unbounded delays.



Real-World Internet Delays (Traceroute)

Traceroute measures per-hop round-trip times by sending three probe packets to each router along the path.

Key observations from sample output:

- Each hop reports three timings (ms); “*” indicates no reply (possible loss or firewall filtering).
- Delays typically increase with geographic distance but can vary due to routing policies and congestion.
- Multiple measurements to the same hop help estimate jitter.

Note: Traceroute relies on the IP Time-to-Live (TTL) field and may be affected by routers that de-prioritize or block probe packets (RFC 1393).



Packet Loss

Loss occurs when a packet arrives at a full buffer (queue) and is dropped.

Consequences:

- The upstream node may **re-transmit** (e.g., TCP’s retransmission mechanisms).
- Excessive loss degrades throughput and triggers congestion-control actions.

Loss probability P_{loss} increases sharply when traffic intensity I approaches 1, especially with finite buffers.



Throughput and Bottleneck Links

Throughput: *Effective data rate (bits/s) achieved between sender and receiver.*

- **Instantaneous throughput** – measured at a specific moment.
- **Average throughput** – measured over a longer interval.

For a path with multiple links, the **end-to-end throughput** is limited by the **bottleneck link**:

$$\text{Throughput} = \min R_1, R_2, \dots, R_n$$

Example

- File size $F = 32$, Mbits, server rate $R_s = 2$, Mbps, client link $R_c = 1$, Mbps .
- Bottleneck = 1, Mbps $\rightarrow$ transfer time = $F/1$, Mbps = 32, s .

Shared Bottleneck Scenario

Four flows share a middle link of capacity $R = 300$, Mbps; each flow's server link $R_s = 50$, Mbps and client link $R_c = 60$, Mbps .

- Per-flow throughput = $\min 50, 60, 300/4 = 50$, Mbps .
- Utilizations: $U_{R_s} = 1$, $U_{R_c} = 50/60 \approx 0.83$, $U_R = 50/(300/4) \approx 0.67$.

☒ Network Devices Overview

Device	Primary Function	OSI Layer	Key Characteristics
Repeater	Regenerates and amplifies physical signals to extend reach	Physical (L1)	No addressing; works on raw bits
Hub	Broadcasts incoming frames to all ports (half-duplex)	Physical (L1)	Single collision domain; no intelligence
Modem	Converts digital $\leftrightarrow$ analog signals for transmission over telephone lines	Physical (L1)	Handles modulation/demodulation

Bridge	Connects two LAN segments; forwards frames based on MAC addresses	Data Link (L2)	Learns MAC table; filters traffic
Switch	Multi-port bridge; directs frames to the correct destination port	Data Link (L2)	Full-duplex, separate collision domains per port
NIC (Network Interface Card)	Provides physical and MAC layer connectivity for an end system	Data Link (L2)	Handles framing, error detection, optional offloading
Router	Routes packets between different IP networks using routing tables	Network (L3)	Performs NAT, fragmentation, QoS
Gateway	Translates between disparate network protocols or architectures	Can span multiple layers	Often implements application-level translation
Firewall	Enforces security policies by permitting/denying traffic	Transport or Application (L4/L7)	Stateful inspection, rule-based filtering

Device Comparison (Expanded)

Feature	Hub	Switch	Router
Function	Broadcasts to all ports	Forwards based on MAC	Routes based on IP
OSI Layer	Physical (L1)	Data Link (L2)	Network (L3)
Collision Domain	Single	One per port	Separate per interface

Bandwidth	Shared among all ports	Dedicated per port	Depends on interface speeds
Intelligence	None (pass-through)	Stores MAC table, makes forwarding decisions	Runs routing protocols, maintains routing table
Typical Use	Small, legacy LANs	Modern LANs, data centers	Inter-network connectivity, Internet edge

Tollbooth Caravan Timing

#	Question	Answer	Brief Reasoning
1	Service time per car	2 s	Given directly.
2	Time for entire caravan (10 cars) to receive service	20 s	$10 \times 2\text{ s} = 20\text{ s}$.
3	Travel time from first-car exit to next tollbooth	50 s	Distance 500 km, speed 10 km/s $\rightarrow 500/10 = 50\text{ s}$.
4	Travel time from last-car exit to next tollbooth	50 s	Same link, same speed; independent of car position.
5	Time until first car is serviced at next tollbooth	68 s	9 cars still in service at first booth ($9 \times 2\text{ s} = 18\text{ s}$) + travel 50 s = 68 s.
6	Can two cars be in service simultaneously at different booths?	No	A car cannot start service at the second booth until all cars have arrived there.
7	Can there be a moment with zero	Yes	After the last car leaves the first

	cars in service (first booth done, second not yet reached)?	booth it must still travel 50 s before any car can be serviced at the second booth.
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One-Hop Transmission Delay

Transmission delay – time to push all L bits onto the link: $d_{\text{trans}} = \frac{L}{R}$.

- **Given:** $L = 8000$ bits, $R = 1 \text{ Mbps} = 1,000,000 \text{ bps}$.

1. Transmission delay

$$d_{\text{trans}} = \frac{8000}{1,000,000} = 0.008 \text{ s}$$

2. Maximum packets per second

$$\text{Packets/sec} = \frac{R}{L} = \frac{1,000,000}{8000} = 125 \text{ packets/s}$$

Queuing Delay (Second Example)

Traffic intensity $I = \frac{L, a}{R}$ (where a is average packet arrival rate).

Queuing delay for $I < 1$: $d_{\text{queue}} = \frac{I, L}{R(1 - I)}$.

- Parameters: $R = 1,900,000 \text{ bps}$, $L = 4700 \text{ bits}$.

1. Does queuing delay vary a lot in practice? → Yes

2. For $a = 35 \text{ packets/s}$

$$I = \frac{4700 \times 35}{1,900,000} = 0.0865$$

$$d_{\text{queue}} = \frac{0.0865 \times 4700}{1,900,000(1 - 0.0865)} = 2.3 \times 10^{-4} \text{ s} = 0.23 \text{ ms}$$

3. For $a = 77 \text{ packets/s}$

$$I = \frac{4700 \times 77}{1,900,000} = 0.1900$$

$$d_{\text{queue}} = \frac{0.1900 \times 4700}{1,900,000(1 - 0.1900)} = 5.8 \times 10^{-4} \text{ s} = 0.58 \text{ ms}$$

4. Buffer evolution (infinite buffer)

- Given queuing delay $0.3815 \text{ ms} = 3.815 \times 10^{-4} \text{ s}$.
- Service rate $= \frac{1}{\text{delay}} \approx \frac{1}{3.815 \times 10^{-4}} \approx 2622 \text{ packets/s}$.
- Arrival rate $a = 35 \text{ packets/s}$ (or 77 depending on scenario).
- After 1 s the buffer empties because service capacity far exceeds arrivals, so **0 packets remain**.

5. Drops with finite buffer (max 627 packets)

- Arriving packets: 613.
- Buffer capacity: 627.
- Since $613 < 627$, **0 packets are dropped**.

End-to-End Delay (Transmission & Propagation) – Third Example

Propagation delay $d_{\text{prop}} = \frac{d}{s}$, with $s \approx 3 \times 10^8 \text{ m/s}$.

- Packet length $L = 12,000 \text{ bits}$.
- Link specifications:

Link	R (Mbps)	Length d (km)
1	100	3
2	1	5000
3	10	1

Calculations

#	Quantity	Formula	Result
1	Transmission delay (Link 1)	$d_{t1} = L/R_1$	0.00012 s
2	Propagation delay (Link 1)	$d_{p1} = (3, \text{km} \times 1000)/3 \times 10^8$	$1.00 \times 10^{-5} \text{ s}$
3	Total delay (Link 1)	$d_1 = d_{t1} + d_{p1}$	0.00013 s

4	Transmission delay (Link 2)	$d_{t2} = L/R_2$	0.012 s
5	Propagation delay (Link 2)	$d_{p2} = (5000, \text{ km} \times 1000)/3 \times 10^8$	0.017 s
6	Total delay (Link 2)	$d_2 = d_{t2} + d_{p2}$	0.029 s
7	Transmission delay (Link 3)	$d_{t3} = L/R_3$	0.0012 s
8	Propagation delay (Link 3)	$d_{p3} = (1, \text{ km} \times 1000)/3 \times 10^8$	$3.33 \times 10^{-6} \text{ s}$
9	Total delay (Link 3)	$d_3 = d_{t3} + d_{p3} \approx 0.0012 \text{ s}$	
10	End-to-end total delay	$d_{\text{total}} = d_1 + d_2 + d_3$	0.03 s

End-to-End Throughput & Bottleneck Links (Re-statement)

- **Shared middle link:** $R = 300 \text{ Mbps}$ (divided equally among 4 flows $\rightarrow 75 \text{ Mbps}$ per flow).
- **Server-to-middle links:** $R_S = 50 \text{ Mbps}$ each.
- **Middle-to-client links:** $R_C = 60 \text{ Mbps}$ each.

#	Question	Answer
1	Maximum achievable end-to-end throughput per flow	50 Mbps (limited by the R_S links).
2	Bottleneck link	R_S (the server-side 50 Mbps link).
3	Utilization of server links R_S	$\frac{50}{50} = 1.00$

4	Utilization of client links R_C	$\frac{50}{60} = 0.83$
5	Utilization of shared middle link R	$\frac{50}{300/4} = 0.67$

Wireshark Overview

Wireshark – a packet capture (pcap) and analysis tool that records a copy of all Ethernet frames transmitted on a network interface.

Layer (OSI / TCP-IP)	Typical Protocol / View in Wireshark
Application	HTTP, FTP, DNS, etc.
Transport	TCP, UDP
Network	IP
Data Link	Ethernet (MAC)
Physical	(not shown directly, but underlying)
Operating System	Captures from NIC drivers